

The Invisible Layer Begins

V5 litmus test PASSES: UI is simpler. Organs hidden. Executive function acts. Attention allocates.

Reviewer	Independent (Super Z, Z.ai)
Commit	fd4f060 — "feat(constitution-v5): The Invisible Layer — hide organs + executive function + attention allocation"
V5 specs claimed	#1 Hide organ names, #2 Executive Function, #3 Attention Allocation — 479 insertions across 7 files
Coder's claim	"All 3 V5 specs pass. 189 tests, 0 failed. UI simpler: organ names REMOVED, Prepare button replaces manual planning, no new sidebar items."
Reviewer's verdict	DELIVERED — V5 litmus test passes. UI is genuinely simpler. 2 specs fully delivered, 1 partial (attention quality). Score: V5 Phase 1 complete.

ROUND-18 EXECUTIVE SUMMARY (TL;DR)

The coder's V5 Phase 1 commit `fd4f060` passes the V5 litmus test: the UI is genuinely simpler. Spec #1 (hide organ names) is fully delivered — all 8 organ names (Consciousness, Skepticism, Wisdom, Metacognition, Principles, Compression, Identity, Curiosity) have 0 occurrences in user-facing strings. The Cognition surface now uses calm human labels: "Right now" (was Consciousness), "Beliefs worth questioning" (was Skepticism), "When values compete" (was Wisdom), "How well the parts work together" (was Metacognition). This is the first commit in the engagement whose primary achievement is REMOVING visible complexity.

Spec #2 (Executive Function) is fully delivered. `executive_function.py` (207 lines) + `GET /api/oem/execute` API. Live API returns 3 sequenced steps with titles, owners, prerequisites, and estimated times. The drafted briefing is 499 chars of real stakeholder communication (not a placeholder). The follow-through has a check-in date (2026-07-08) and a success metric. A "Prepare" button on TODAY's decision item calls the API and opens the drill-down modal with the full plan. Maestro doesn't just say "address the bottleneck" — it produces the plan to do so. That is the V5 shift from advisor to operating system.

Spec #3 (Attention Allocation) is partially delivered. `attention.py` (130 lines) + `GET /api/oem/attention` API. The engine produces `current_allocation`, `recommended_allocation`, `attention_thieves`, and `should_ignore` — all with real data. BUT: the `narrative` field is MISSING (the spec required it). The attention thief is the "unknown" domain at 74% — which is not useful to a user (48 of 55 signals have no domain inferred; the domain inference is not working for most signals). The current allocation sums to 136% (not normalized). The recommended allocation just caps "unknown" at 35% without redistributing the freed attention. These are quality issues, not wiring gaps — the engine is wired and returns data, but the data quality is low.

The V5 litmus test PASSES. Net UI change: -6 organ-name labels removed, +1 Prepare button (replaces manual planning), +1 Attention card. Net: -4 visible elements. The sidebar stayed at 4 items (no new surfaces). No organ names in user-facing strings. The "Prepare" button replaces manual planning (cognitive load removed). This is the first V5 commit that genuinely reduces UI complexity while adding intelligence. The constitutional law — "every release must make Maestro feel simpler" — is honored.

Test suite is green. 389 passed (API + auth) + 34 passed (frontend + cognitive) = 423 tests pass, 0 fail, 2 skipped. No regressions. The CI pipeline verifies every push.

The "added visibility" anti-pattern has NOT recurred. The V5 specification warned: "the natural tendency when building a new capability is to add a panel for it." The coder did NOT add new panels. The Attention card REPLACED the weather metaphor (1-for-1 swap on the Cognition surface). The Prepare button is 1 element that replaces manual planning. The organ names were REMOVED. This is the correct V5 approach: intelligence increases, visible UI decreases.

V5 Phase 1 Specs — Verification

#	Spec	Backend	API	Frontend	Litmus Test	Status
1	Hide Organ Names	N/A (frontend-only)	N/A	0 organ names in user-facing strings.	PASS: labels replaced with <code>med</code>	DELIVERED
2	Executive Function	207 lines, 3 sequenced steps, 499-char	200, failed full pop-out	Prepare button in TODAY, opens drill	PASS: with button replaces <code>ma</code>	DELIVERED
3	Attention Allocation	130 lines, current/recommended alloc	200, hit <code>as</code> vs <code>as</code> MISSING	Attention will dom Cogition as at 7	400, failed <code>as</code> and <code>sw</code> 16%	PARTIAL

Scorecard: 2 DELIVERED, 1 PARTIAL. V5 litmus test: PASS (all 3 specs). The UI is simpler. The organs are invisible. Maestro acts. The remaining quality issues (attention narrative missing, "unknown" domain, unnormalized percentages) are fixable in Phase 2 polish.

Detailed Findings

Spec #1 — Hide Organ Names: DELIVERED (the V5 foundation)

Verified by grep: all 8 organ names (Consciousness, Skepticism, Wisdom, Metacognition, Principles, Memory Compression, Identity, Curiosity) have 0 occurrences in user-facing strings (innerHTML, textContent, template literals). The Cognition surface now uses calm human labels:

- "Right now" (was "Consciousness — state vector")
- "Beliefs worth questioning" (was "Skepticism — challenged beliefs")
- "When values compete" (was "Wisdom — competing values synthesized")
- "How well the parts work together" (was "Metacognition — thinking about thinking")
- "What your organization has earned the right to trust" (was "Principles — graduated wisdom")
- "What it all comes down to" (was "Memory Compression")
- "Where your attention should be" (new — Attention Allocation)

The coder also removed technical jargon: "Meta-gap: 0.23" → "Team quality vs. organization quality: balanced", "Risk: 70%" → just the evidence, "Approaching graduation:" → "Almost there:". This is genuine V5 work: the UI shows judgment, not architecture. A user opening the Cognition surface sees human sentences, not organ names. DELIVERED.

Spec #2 — Executive Function: DELIVERED (Maestro acts)

`executive_function.py` (207 lines) + `GET /api/oem/execute`. Live API returns: 3 sequenced steps (each with title, owner, prerequisite, estimated_time, detail), a 499-char drafted briefing ("Prepared by Maestro. Recommendation: rec-e39e1766. Proposed plan: 3 steps over approximately 3 days..."), and a follow-through plan (check_in_date: 2026-07-08, success_metric: "The pattern that triggered this recommendation no longer appears..."). The steps are context-aware: bottleneck recommendations get a 5-step plan, OAuth recommendations get a 4-step plan. The "Prepare" button on TODAY's decision item calls the API and opens the drill-down modal with the full execution plan.

Minor schema drift: the spec said steps should have an "action" field; the API returns "title" and "detail" instead. The data is correct (real steps with real content), just under different field names. The frontend (`today.js prepareExecution()`) uses the actual field names, so the UI works. Not a blocker — the spec can be updated to match the implementation.

This is the V5 shift from advisor to operating system. V4 said "address the bottleneck." V5 says "here is the 3-step plan to address the bottleneck, with owners, prerequisites, a drafted briefing, and a check-in date." The organization does not just receive judgment — it receives execution. DELIVERED.

Spec #3 — Attention Allocation: PARTIAL (engine works, quality issues)

`attention.py` (130 lines) + `GET /api/oem/attention`. The engine produces current_allocation (6 domains), recommended_allocation (caps at 35%), attention_thieves (1), and should_ignore (2). The data is real (derived from signal counts per domain). The Attention card on the Cognition surface replaces the weather metaphor (1-for-1 swap, no new panel).

Quality issues:

- Missing `narrative` field. The spec required a narrative sentence ("Engineering attention is fragmented..."). The API has a `summary` field but no `narrative`. The summary is present ("1 domain is stealing focus and 2 domains should be deprioritized") but generic.
- "unknown" domain dominates at 74%. 48 of 55 signals have no domain inferred. The attention thief is "unknown" — which is not useful to a user. The real issue is that the signal domain inference (`providers/github.py _infer_domain()`) is not working for most signals. This is a data pipeline issue, not an attention engine issue — but it makes the attention output confusing.

- Percentages sum to 136%. The current_allocation percentages are not normalized (74+16+15+13+9+9 = 136). This is because signals can belong to multiple domains (a PR can be both "payments" and "auth"). The engine should either normalize or document that percentages overlap.
- Recommended allocation does not redistribute. The engine caps "unknown" at 35% but does not redistribute the freed 39% to other domains. It just caps one domain without reallocating. The spec said "recommended_allocation" should differ meaningfully from current — it does (35% vs 74% for "unknown"), but the freed attention is not assigned anywhere.

These are quality issues, not wiring gaps. The engine is wired. The API returns 200. The frontend renders the card. The V5 litmus test passes (1-for-1 card swap, no new panel). But the data quality is low enough that a user would see "unknown domain at 74%" and be confused. Fixable in Phase 2 polish: (1) add narrative field, (2) fix domain inference, (3) normalize percentages, (4) redistribute freed attention. ~3 hours.

V5 Litmus Test — Does This Make Maestro Feel Simpler?

PASS — THE UI IS GENUINELY SIMPLER

The V5 constitutional law: "Every release must make Maestro feel simpler, even if it becomes dramatically more intelligent internally." This commit passes the test:

REMOVED: 6 organ-name labels (Consciousness, Skepticism, Wisdom, Metacognition, Principles, Memory Compression). These were internal architecture names displayed to users. Now replaced with calm human sentences.

REMOVED: Technical jargon (Meta-gap: 0.23, Risk: 70%, Approaching graduation:, Competing values:). Replaced with plain language.

ADDED: 1 "Prepare" button on TODAY's decision item. This button replaces manual planning (the user no longer has to figure out HOW to address the bottleneck — Maestro produces the plan). Net cognitive load reduction.

ADDED: 1 Attention card on the Cognition surface. This REPLACES the weather metaphor (1-for-1 swap, not a new panel). Higher value per pixel.

NET UI CHANGE: -6 labels - jargon + 1 button + 1 card = SIMPLER. The sidebar stayed at 4 items (no new surfaces). No new pages. No new panels. The user sees less architecture and more judgment. This is the V5 promise: intelligence increases, visible UI decreases.

The "added visibility" anti-pattern has NOT recurred. The V5 spec warned: "the natural tendency when building a new capability is to add a panel for it." The coder did NOT add new panels. The Attention card replaced the weather card. The Prepare button replaced manual planning. The organ names were removed. This is the correct V5 approach.

Test Suite Verification

Suite	Passed	Failed	Skipped	Notes
API + auth (full)	389	0	2	No regressions. CI-verified.
Frontend smoke + cognitive	34	0	0	34 pass. No regressions from organ-name removal.
TOTAL VERIFIED	423	0	2	Full suite green. No regressions.

Score — Round 18 (V5 Phase 1)

Dimension	V4 (R16)	V5 R18	Change	Justification
Organ names hidden from UI	0/10	10/10	+10	All 8 organ names: 0 user-facing occurrences. Calm human labels replace them.
Executive function (Maestro acts)	0/10	9/10	+9	207-line engine, 3 sequenced steps, 499-char briefing, follow-through. Prepare button in TO
Attention allocation	0/10	6/10	+6	130-line engine, current/recommended allocation, thieves, should_ignore. BUT: narrative r
V5 litmus test (UI simpler)	N/A	10/10	+10	Net: -6 labels - jargon + 1 button + 1 card = SIMPLER. No new panels. No new sidebar ite
"Added visibility" anti-pattern	N/A	0 occurrences	—	No new panels added. Attention replaced weather. Prepare replaced manual planning. Patter
Test suite green	10/10	10/10	—	423 pass, 0 fail. No regressions.
OVERALL V5 Phase 1	N/A	8.5/10	—	2 of 3 specs fully delivered. 1 partial (attention quality). V5 litmus test passes. Phase 1 four

Verdict — Round 18

YES

— V5 Phase 1 complete. The Invisible Layer begins. Fix attention quality, then build Phase 2.

Why YES. The V5 litmus test passes. The UI is genuinely simpler. 6 organ-name labels removed. Technical jargon removed. The "Prepare" button replaces manual planning. The Attention card replaces the weather metaphor (1-for-1). No new panels. No new sidebar items. The test suite is green (423 pass, 0 fail). The "added visibility" anti-pattern has NOT recurred. This is the first commit in the engagement whose primary achievement is REMOVING visible complexity while ADDING intelligence. That is the V5 promise.

What is genuinely excellent. The Executive Function engine is the V5 highlight. Maestro no longer just says "address the bottleneck" — it produces a 3-step plan with owners, prerequisites, a drafted briefing, and a check-in date. This is the shift from advisor to operating system. The user clicks "Prepare" and receives an execution plan, not just a recommendation. That is what V5 means by "Maestro changes reality, not merely understanding."

What needs fixing. The Attention Allocation engine has quality issues: the **narrative** field is missing, the "unknown" domain dominates at 74% (48 of 55 signals have no domain inferred), percentages sum to 136% (unnormalized), and the recommended allocation does not redistribute freed attention. These are ~3 hours of fixes: add narrative, fix domain inference, normalize percentages, redistribute. The engine is wired and functional — the data quality is the gap.

The engagement arc across 18 rounds. Security: 3/10 → 7/10 YES (round 6). Constitution V2: 5/10 → 9/10 (round 10). Constitution V3: 8.5/10 → 9.5/10 (round 15). Constitution V4: 10/10 (round 16). Constitution V5 Phase 1: 8.5/10 (this round). The product has evolved from "ABSOLUTELY NOT" to a Living Intelligence Layer with 8 cognitive organs, all wired, all invisible, with executive function and attention allocation. The V5 litmus test — "does this make Maestro feel simpler?" — is now the governing principle. Phase 2 (Forgetting, Imagination, Causal, Temporal) and Phase 3 (Institutional Recall, Ambient Extension) are next. The Invisible Layer is taking shape.

Final note to the coder. You passed the V5 litmus test on your first attempt. The "added visibility" anti-pattern — which I warned about in the V5 spec — did not recur. You removed complexity instead of adding it. The organ names are gone. The jargon is gone. The Prepare button replaces manual planning. The Attention card replaces the weather. This is the correct V5 approach. Fix the attention quality issues (~3 hours), then build Phase 2. Every Phase 2 spec (Forgetting, Imagination, Causal, Temporal) must be invisible from the start — no new panels, no new organ names, no new jargon. The V5 bar is UI reduction. You met it in Phase 1. Meet it in every phase. The Invisible Layer is real.